DIVISION 23

HEATING, VENTILATING, AND AIR-CONDITIONING

SECTION 23 05 13

COMMON ELECTRICAL REQUIREMENTS FOR HVAC EQUIPMENT

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Architectural and Structural Specification sections apply to work of this section.

B. Separate electrical components and materials required for field installation and electrical connections are specified in Division 26.

C. Refer to Division 26 “ELECTRICAL” for Motor Control Centers, Motor Control Panels, Cables, Circuit Protection and Switches, etc.

1.02 SCOPE OF WORK

A. This Section specifies the basic requirements for electrical components which are an integral part of packaged mechanical equipment. These components include, but are not limited to factory installed motors, starters, and disconnect switches furnished as an integral part of packaged mechanical equipment.

B. Specific electrical requirements (i.e. horsepower and electrical characteristics) for mechanical equipment are specified within the individual equipment specification sections, or are scheduled on the Drawings.

1.03 CODES AND STANDARDS

A. Electrical components of mechanical equipment to conform to the following codes and standards:

NEMA Standards MG1: Motors and Generators

NEMA Standards ICS 2: Industrial Control Devices, Controllers, and Assemblies

NEMA Standard 250: Enclosed Switches

National Electrical Code (NFPA 70)

1.04 MANUFACTURERS

A. Please Refer to Approved List of Manufacturers.

1.05 SUBMITTALS

A. No separate submittal is required. Submit product data for motors, starters, and other electrical components with submittal data required for the equipment for which it serves, as required by the individual equipment specification sections.

B. Compliance statement clause by clause full and detailed compliance statement of the specification shall be submitted where a clear compliance is indicated as follows:

a. “Yes” interprets full unconditional compliance.
b. “No” interprets a deviation (to be clearly indicated as remark or note in the dedicated column).
c. “Not Applicable” where the case is.

It should be noted that other comments like “noted” or “partial compliance” are unacceptable and will be interpreted as a non compliance.

1.06 QUALITY ASSURANCE

A. Electrical components and materials to be according to specified standards .

PART 2 - PRODUCTS

2.01 CONDUITS, WIRES AND CABLES

A. Conduits

1. SIZE: minimum 20 mm unless otherwise specified.
2. STEEL CONDUIT: welded, drawn, heavy gauge, to BS 4568 Part 1 or approved equal, hot dip galvanized internally and externally and threaded both ends.
3. FITTINGS: threaded, hot dip galvanized or cadmium plated malleable iron.
4. FITTINGS to be specifically designed for size and type of conduit.

B. Wires and Cables

1. Power Wires and Cables: 600/1000 V grade, to IEC 502 or BS 6346. Single conductor wires and multicore cables to have high conductivity tinned copper wire conductors insulated with XLPE compound, with additional PVC sheath for multicore cables.
2. Control Wires and Cables: conductors to be tinned annealed copper, minimum area 1.5 sq mm. Insulation to be moisture resistant flame retardant PVC compound. Wires and cables to be rated at 450/750 V to BS 6004. Multicore cables for control and signalling to be polyethyleme insulated copper conductors, PVC sheathed, rated at 600/1000 V to BS 6346 as indicated on the drawings.
3. Control Wires and Cables: provide special heat resistant insulation or corrosion resistant sheath where required.
4. Armoured Cables: armour to be single layer of galvanized steel wire under the PVC sheath for the multicore cables.
5. Connections TO MOTORS: single conductor wires pulled inside conduits, or multicore cables armoured or non armoured and fixed on cable trays or supports. Industrial flexible rubber cables to be installed for vibrating equipments (i.e. process pump and fan motors through intermediate connecting panel, make-up fresh air, extract fan, AHU, FAHU, etc.).

6. SPECIAL CONDITIONS: where required high temperature resistant cables, silicone rubber, cross linked polyethylene, MICC, or approved equal, corrosion resistant sheath is to be provided.

2.02 ELECTRIC MOTORS

A. Motors to be supplied by driven equipment manufacturer, to be as specified for equipment concerned and specifically supplied for available supply voltage and frequency.

B. Motors 1/2 horsepower and under to be single phase and over 1/2 horsepower to be three phase. Motors to be totally enclosed, fan cooled type, unless otherwise specified.

C. Motors to have Class F insulation with 80 deg. C continuous temperature rise above average ambient temperature of 50 deg. C, unless otherwise specified.

D. Motors that will operate outdoors are to have Class F insulation. Motors used for kitchen or smoke control to have class H insulation.

E. Motors operating in ambient temperatures 50 deg. C to be tropicalized and derated for satisfactory operation.

F. Motors to be rated for continuous operation with service factor of 1.15 and efficiency for all motors to be premium (IE3) or super premium based on the available size.

G. POWER to be adequate to operate driven equipment without motor overload under all operating conditions and loads and throughout capacity range of equipment. Motor to be capable of delivering full rated output when operating at voltage deviating by 5% from rated voltage at rated frequency.

H. Starting and Torque Characteristics to be as required by driven equipment.

I. Speed to be as specified for equipment concerned.

J. Conduit Terminal Box on Motor to be approved model for type of motor enclosure. Motor windings to be connected to terminals in terminal box at factory. One additional earthling terminal to be connected to motor frame.

K. Motor Base to be adjustable where motors are directly connected to driven equipment, unless otherwise specified. Motors connected to equipment through V-belt drive to have adjustable sliding base. Fractional horsepower motors to have slotted mounting holes in base

2.03 COMPONENTS AND ACCESSORIES

A. Starters

1. Starters for three phase motors to be magnetic type to automatically disconnect motor from power supply in case of supply failure, excessive voltage drop, over current and lack of balance in phases. Overload trips to be provided for three phases.

2. Motor Data: obtain from equipment supplier before ordering any motor starter, or check motor nameplate for full load current rating and allowable temperature rise in order to select proper overload thermal element for motor starter.

3. Short Circuit Protection Device fitted to starter to be independent of controller and overload protection.

4. CONTROL VOLTAGE for starters and control circuits is not to exceed 110 V.

5. Step Down Control Circuit Transformers: two winding isolating type.

6. Control Circuit Protection: use high rupturing capacity fuses or circuit breakers.

7. Auxiliary Supply for controls, other than from main power circuit, to be effectively isolated by auxiliary contacts on main isolator.

8. Control Devices on starters to be as follows unless otherwise indicated or required by driven equipment: start stop push buttons, one red pilot light for "running", one group pilot light for "stopped" and one reset push button.

9. Starter Type A for single phase motors not exceeding 1/2 HP to be surface or flush mounted, manual two pole toggle type, for non reversing across the line starting, fitted with one overload element.

10. Starter Type B for three phase motors not exceeding 10 HP to be direct on line, non reversing, magnetic type, with manual reset, 3 pole overload relay and low voltage protection, unless otherwise required by local regulations.

11. Starter Type C for three phase motors over 10 HP, but not exceeding 50 HP, to be automatic star delta magnetic non reversing type or softstarter as applicable, with 3 pole overload relay and low voltage protection, unless otherwise required by local regulations.

12. Starter Type D for three phase motors over 50 HP to be driven by softstarter or variable speed drive, as applicable, with 3 pole overload relay, low voltage protection, unless otherwise required by local regulations.

13. Individually Mounted Starters to be totally enclosed in sheet steel enclosure with baked enamel finish. Design is to suit location and application. It is to be impossible to open enclosure door unless isolator is in open position.

14. Nameplates: starters and controls to have engraved nameplates identifying system or defining its function.

15. Contactors are to comply with IEC 60947-4, utilization category AC3, and/or UL Standard 1008, and be 3-phase, 4-pole, magnetic type, 600 V rating, capable of interrupting at least ten times rated current inductive or non-inductive loads under normal service conditions and are to have replaceable main arcing contacts and arc quenching devices. Contactors are to withstand, without welding or burning of contacts, an inrush current of 20 times normal rating for 4 seconds upon closing and are to be capable of closing on the heaviest short-circuit of the system and withstand the short-circuit for period required by upstream short-circuit protective device to operate. Three N.O. and three N.C. spare contacts are to be provided on each contactor.

16. STARTER COORDINATION: motor starter devices shall be of type 2 coordination to IEC 60947-4-1.

B. Switch Disconnectors

1. Switch Disconnectors: non fusible, single throw type, with AC-23 utilization category to IEC 60408 & IEC 947-3 for AC motors, housed in separate metallic enclosure, with arc quenching devices on each pole capable of interrupting at least six times its rated current. They are to simultaneously interrupt power supply to all line conductors, any neutral and control circuits.

2. Switch Disconnectors for single phase fractional horsepower motors to be single pole, dolly operated type, rated 15/20 A at 250 V AC, quick make, quick break, with silver alloy contacts, flush or surface mounted to suit application.

3. Operating Mechanism to be quick make quick break type, with external operating handle mechanically interlocked with enclosure cover to necessitate disconnecting switch to be in OFF position for access to inside of enclosure. Means are to be provided for by passing interlocks. Position of isolating switch to be clearly indicated on cover.

4. Enclosure to be general purpose type, unless otherwise indicated or required, with provision for locking operating handle in OPEN and CLOSED positions.

5. Enclosures where indicated or required by location to be weatherproof totally sealed water and dust proof type.

C. Combination Starters Switch Disconnectors

1. Components to comprise magnetic starter, switch disconnector and short circuit protection devices required by the Standards, in approved sheet metal enclosure to suit application.

2. Switch Disconnector Operating Mechanism: quick make, quick break, with external operating handle mechanically interlocked with enclosure cover necessitating disconnecting switch to be in OFF position for access to inside of enclosure. Means are to be provided for by passing interlocks. Position of isolating switch to be clearly indicated on cover.

3. Short Circuit Protection Gear to be HRC fused cartridges or moulded case circuit breakers of appropriate current rupturing capacity. Switch disconnectors are not required if circuit breakers are used for the short circuit protection. In this case the circuit breaker will perform the disconnection function.

4. Operation of Circuit Breaker to be possible from outside of enclosure. Position of breaker ON/OFF/TRIPPED to be clearly indicated by position of handle.

D. Push Buttons

1. Push Buttons to be one unit momentary contact START/STOP with normally open or normally closed contacts as required by wiring diagrams and with lockout attachments. Heads to be colour coded and STOP button to be protected. Push buttons controlling one piece of equipment to be housed in separate enclosure.

E. Relays

1. Relays to be multiple with normally open or normally closed contacts, electrically operated at 110 V maximum, and magnetically held. Contacts to be double break, silvered type,

interchangeable from normally open to normally closed without additional parts. Relays are to be rated at 10 A, 600 V.

F. Circuit Breakers

1. Circuit Breakers: thermal magnetic type, with moulded case, manually operated for normal switching functions and automatically operated under overload and short circuit conditions.

2. Circuit Breakers to give positive trip free operation on abnormal overloads, with quick make quick break contacts under both manual and automatic operation. Stationary and movable contacts to be non-welding silver alloy adequately protected with effective and rapid arc interruption.

3. Motor Circuit Protector: moulded case, magnetic break type with adjustable instantaneous setting suitable for motor protection.

4. Moulded Case Switch: similar to circuit breakers but without overload/short circuit protection. Short time rating of switches is to be not less than the short circuit current at switch location for 3 cycles. Switches are to have a suitable self-override instantaneous protection and to be in compliance with IEC 947-3.

5. Branch Circuit Breakers to be 100 A frame size, unless otherwise shown.

6. Main Circuit Breakers to be 100 A frame size or larger as shown on the Drawings.

7. Circuit Breakers and Motor Circuit Protectors are to be in compliance with IEC 947-2, utilization category B, sequence II (service capacity) for motor control centers and sequence III (ultimate capacity) for motor control panels and combination starters unless otherwise indicated on the Drawings.

8. Breakers OF 225 AMP FRAME SIZE and larger to have interchangeable trip units and adjustable instantaneous trips unless otherwise shown.

9. MAIN INCOMING BREAKERS / SWITCHES to be equipped to provide earth fault, under voltage and phase sequence protection through shunt trip coil. Earth fault detection and interruption to be time coordinated with those of main incoming breaker on main distribution board.

10. Multiple Pole Breakers to have single handle mechanism. Each pole to have inverse time delay thermal overcurrent trip element and magnetic instantaneous overcurrent trip element for simultaneous tripping of all poles.

11. Trip Elements to be ambient temperature compensated type.

G. MOTOR CONTROL CENTER

1. Type: totally enclosed, IP 42 for indoor installation and IP 55 for outdoor installation and in wet areas (e.g pump rooms), freestanding sectionalized type, modular, compartmented, form 3 to IEC 61439 with separate withdrawable section for each motor starter and separate fixed section for main incoming breaker / switch and each outgoing feeder circuit breaker. Sections to be designed to allow other sections to be easily added or removed.

2. Construction: 2 mm thick sheet steel, adequately reinforced and braced for maximum rigidity, sand blasted, rust inhibited after fabrication and sprayed with one coat primer and two coats enamel internally and externally.

3. Components: motor control center to include the following:

   a. main incoming circuit breaker or isolating switch as shown on the Drawings for terminating incoming supply cables and isolating the busbar system
   b. main copper horizontal full length busbars rated as main incoming circuit breaker or as shown on the Drawings
   c. branch copper vertical full height busbars of adequate capacity to distribute power to each circuit breaker and starter served
   d. neutral copper busbar rated at half capacity of main busbar and distributed throughout whole motor control centre
   e. earth copper busbars minimum 25 x 6 mm extending full length of motor control centre
   f. one voltmeter with range 0-415 V
   g. ammeters on main supply with necessary current transformers
   h. starters, circuit breakers, push buttons, indicating lights, switches, relays, contactors and accessories as shown on the Drawings
   i. interconnecting and interlock wiring.

4. Wiring: the motor control centre wiring to be class II type B to NEMA ICS.

5. Busbars to be adequately isolated and braced to sustain maximum possible short circuit current.

6. Compartment Doors to be interlocked so that isolators or breakers must be in OFF position before door can be opened.

7. Spare Positions: fully equipped cells ready for connection to motors are to be provided in adequate number.

8. Space Positions: fully equipped cells ready to receive control unit are to be provided in adequate number. Space positions to have blank cover plates.

9. Starters, Switches, other components and electrical devices to be clearly labeled in English as to number and function, with incised letters on black Bakelite with white laminated core. Labels to be permanently fixed under each component.

10. Incoming Line Connections to be made with solder less, terminal four bolt type clamps.

11. Labels: starters, switches, electrical devices and accessories to be clearly labeled in English as to function and number. Labels to be permanently fixed under each component.

12. Schematic and Wiring Diagrams to be firmly fixed within motor control center, showing each component and cross- referenced with component labels.

13. Submit for Approval electrical schematic diagram of whole installation, suggested layout of motor control center, interior wiring details and complete technical information on all proposed components, prior to fabrication or purchase.

H. Motor Control Panels

1. Type: wall mounted or unit mounted, lockable type, IP 42 for indoor installation.

2. Construction: minimum 1.5 mm thick hot-dip galvanized steel sheet, finished with one coat etch primer and one coat stove enamel internally and externally.

3. Panels Installed Outdoors and in Wet Areas to have weatherproof totally sealed water and dustproof IP 55 enclosures.

4. Components: panels are to contain necessary breakers, starters, push button switches, selector switches, relays, indicating lights, interconnecting and interlock wiring and all devices and accessories required for automatic or manual operation of equipment as specified under equipment concerned.

5. Labels: starters, switches, electrical devices and accessories to be clearly labelled in English as to function and number. Labels to be permanently fixed under each component.

6. Schematic and Wiring Diagrams to be mounted in permanent approved manner on inside of panel door. Diagrams are to show each component cross referenced with component labels.

I. Control Switches

1. Float Switch: level operated, heavy duty, bracket mounted type, suitable for application in open tanks, complete with 178 mm spun copper float, brass rod, two stops, floor mounting stand, lever and counterweight. Switch to have oil tight and dust tight enclosure and 2 pole double throw silver contacts that open on liquid rise.

2. Float Switch to be as manufactured by Square D, Type BW 3, or approved equal.

3. Pressure Switch: industrial, heavy duty, bellows actuated type, suitable for water service, with contacts to close on falling pressure. Range to be 0.1 to 8 kg/cm2. Switch to be good for 1720 kPa

4. operating pressure and to have 6 mm pipe tap bottom connection. It is to have oil tight and dust tight enclosure, single pole double throw contacts and setting adjustment.

5. Pressure Switch to be as manufactured by Square D Company Type ACW-1 or approved equal.

6. Low Suction Pressure Switch: industrial, sensitive, low range, diaphragm actuated type, suitable for water service, with range of 2 to 20 kPa of falling pressure, preset at factory to 3 kPa. Switch to be good for 690 kPa operating pressure and to have 6 mm pipe tap bottom connection. It is to have oil tight and dust tight enclosure, single pole double throw contacts, range adjustment knob, sealing cap and range locking nut.

7. Low Suction Pressure Switch to be as manufactured by Square D Company Type AMW-1 or approved equal.

2.04 VARIABLE FREQUENCY DRIVE

A. Furnish complete variable frequency drives as specified herein for the fans and pumps designated to be variable speed. All standards and optional features shall be included within the

VFD enclosure, unless otherwise specified. VFD enclosure shall be IP 54 with inlet air filters. Drive shall be mounted on a floor stand. Drive to be Active front end- low harmonic drive (THD<5 %), unless otherwise specified.

B. The VFD shall convert three-phase, 50 hertz utility power to adjustable voltage and frequency, three phase power for stepless motor speed control from 10% to 100% of the motor's 50 hertz speed. Input voltage shall be as specified on the drawing schedules.

C. The VFD shall include a converter and an inverter section. The converter section shall convert fixed frequency and voltage AC utility power to DC voltage. All VFDs shall include input line reactors or an isolation transformer as required.

D. The inverter section of the VFD shall invert the DC voltage into a quality output wave-form, with adjustable voltage and frequency for stepless motor speed control. The VFD shall maintain a constant V/Hz ratio.

E. The VFD and options shall be tested to ANSI/UL.508. The complete drive, including all specified options, shall be listed by a nationally recognized testing agency such as UL or ETL.

F. Power line noise shall be limited to a voltage distortion factor and line notch as defined in IEEE 519 "Guide for Harmonic Control and Reactive Compensation of Static Power Converters". The total voltage distortion shall not exceed 5 percent.

G. The VFD shall not emit radiated RFI in excess of the limitations set forth in the FCC Rules and Regulations, FCC Part 15 for Class A computing devices. The VFD shall carry a FCC compliance label. PWM type drives shall include RFI filters.

H. Motor noise as a result of the VFD shall be limited to 3 db over across the line operation, measured at 1 meter from the motor's centre line.

I. The VFD's full load amp rating shall meet or exceed NFPA 70, NEC Table 430-150.

J. Protective Features:

1. Individual motor overload protection for each motor controlled.
2. Protection against input power under-voltage, over-voltage, and phase loss.
3. Protection against output current overload and instantaneous over-current.
4. Protection against over-temperature within the VFD enclosure.
5. Protection against over-voltage on the DC bus.
6. Protect VFD from sustained power of phase loss. Under-voltage trip activates automatically when line voltage drops more than 10 percent below rated input voltage.
7. Automatically reset faults due to under-voltage, over-voltage, phase loss, or over-temperature.
8. Protection against output short circuit and motor winding shorting to case faults, as defined by UL 508.
9. Status lights or digital for indication of individual fault conditions.
10. Controller capable of operating without a motor or any other equipment connected to the drive output to facilitate start-up and troubleshooting.

11. Input line reactors shall be provided to minimize harmonics introduced to the AC line and to provide additional protection to AC line transients.

K. Interface Features:

1. Door mounted Hand/Off/Auto selector switch to start and stop the VFD. In the Auto position, the VFD will start/stop from the controller. In the Hand position, the VFD will run regardless of the remote contact position.
2. Manual speed control capability.
3. Local/Remote selector switch. In the Remote position, motor speed is determined by the follower signal. In the local position, motor speed is determined by the manual speed control.
4. Power/On light to indicate that the VFD has tripped on a fault condition.
5. Digital meter with selector switch to indicate percent speed and percent load.
6. A set of form-C, dry contacts to indicate when the VFD is in the Run mode.
7. A set of form-C, dry contacts to indicate when the VFD is in the Fault mode.
8. A 0 to 10 Vdc output signal to vary in direct proportion to the controller's speed.
9. VFD to have terminal strip and N.C. safety contacts such as freezstats, smoke alarms, etc. VFD to safety shut down in drive or bypass mode when contacts open.

L. Adjustments:

1. Maximum speed, adjustable 15 to 100 percent base speed.
2. Acceleration time, adjustable 3 to 60 seconds.
3. Deceleration time, adjustable 3 to 60 seconds with override circuit to prevent nuisance trips if decal time is set too short.
4. Current limit, adjustable 0 to 105 percent.
5. Overload trip set point.
6. Offset and gain to condition the input speed signal.

M. Service Conditions:

1. Ambient temperature, 0 to 50 degrees C.
2. 0 to 95 percent relative humidity, non condensing.
3. Elevation to 3,300 feet (1,000 meters) without derating.
4. AC line voltage variation, -10 to +10 percent of nominal.

N. Special Features: The following special features shall be included in the VFD enclosure. The unit shall maintain its UL to ETL listing.

1. Manual by-pass shall provide all the circuitry necessary to transfer the motor from the VFD to the power line, or from the line to the controller. The by-pass circuitry shall be mounted in a separate section section of the VFD enclosure. Motor overload protection shall be provided in both drive and by-pass modes.

2. A door interlocked, pad locakble drive disconnect switch shall be provided to disconnect power from the VFD only (the disconnect shall be clearly marked as such).
3. A motor circuit protector, MCP, shall be provided as a means of disconnecting all power to both the VFD and by-pass circuits as well as provided short circuit and locked rotor protection to the motor while in the by-pass mode.
4. The disconnect and by-pass functions may be accomplished via disconnects, contractors and overloads, or with a four position Drive/Off/Line/Test switch with motor starter and by-pass.

O. Quality Assurance:

1. To ensure quality and minimize infantile failures at the job site, the complete VFD shall be tested by the manufacturer. The VFD shall operate a dynamometer at full load and the load and speed shall be cycled during the test.
2. All optional features shall be functionally tested at the factory for proper operation.

P. Submittals:

1. Submit manufacturer's performance data including dimensional drawings, customer connection drawings, power circuit diagrams, installation and maintenance manuals, warranty description, VFDs FLA rating, certification agency file numbers and catalogue information.

PART 3 - EXECUTION (NOT APPLICABLE)

END OF SECTION 23 05 13

SECTION 23 21 23

HYDRONIC PUMPS

PART 1 – GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.02 SUMMARY

A. This Section includes the following categories of HVAC pumps for hydronic systems:
   1. End suction centrifugal pump.
   2. In-line circulators.
   3. Double-suction pumps

B. Related Sections: The following Sections contain requirements that relate to this Section:
   1. Division 23 Section 230519 "Meters and Gages for HVAC Piping" for thermometers and pressure gages, connector plugs, and devices.
   2. Division 23 Section 230513 "Common Electrical Requirements for HVAC Equipment" for pump motors.
   3. Division 25 for inertia pads, isolation pads, spring supports, spring hangers, and flexible pipe connectors.
   4. Division 25 for interlock wiring between pumps, and between pumps and field-installed control devices.
   5. Division 23 and 26 Sections for power-supply wiring, field-installed disconnects, required electrical devices, and motor controllers.
   6. Division 23 & 26 Sections "Variable Frequency Drivers".
   7. Division 26 - (Section 26 24 19) - MOTOR CONTROL CENTERS.

1.03 PERFORMANCE REQUIREMENTS

A. Pump Pressure Ratings: At least equal to system's maximum operating pressure at point where installed, but not less than specified.
B. The contractor shall provide pumps complete with internal polishing for impeller and casing to achieve the highest possible pump efficiency.

1.04 SUBMITTALS

A. General: Submit each item in this Article according to the Conditions of the Contract and Division 1 Specification Sections.

B. Product data including certified performance curves and rated capacities of selected models, weights (shipping, installed, and operating), furnished specialties, and accessories. Indicate pump's operating point on curves.

C. Hydraulic modelling for all pumping systems shall be carried out by contractor using a reputable well known software and submitted to the Engineer for approval.

D. Shop drawings showing pump layout and connections. Include setting drawings with templates, directions for installation of foundation and anchor bolts, and other anchorages.

E. Wiring diagrams detailing wiring for power, signal, and control systems and differentiating between manufacturer-installed wiring and field-installed wiring.

F. Maintenance data for pumps to include in the operation and maintenance manual specified in Division 1. Include startup instructions.

G. Variable frequency drives.

H. Compliance statement clause by clause full and detailed compliance statement of the specification shall be submitted where a clear compliance is indicated as follows:

a. “Yes” interprets full unconditional compliance.
b. “No” interprets a deviation (to be clearly indicated as remark or note in the dedicated column).
c. “Not Applicable” where the case is.

It should be noted that other comments like “noted” or “partial compliance” are unacceptable and will be interpreted as a non compliance.

1.05 QUALITY ASSURANCE

A. Regulatory Requirements: Comply with provisions of the following:

1. ASME B31.9 "Building Services Piping" for piping materials and installation.
2. Hydraulic Institute's "Standards for Centrifugal, Rotary & Reciprocating Pumps" for pump design, manufacture, testing, and installation.
3. UL 778 "Standard for Motor Operated Water Pumps" for construction requirements. Include UL listing and labeling.
4. NEMA MG 1 "Standard for Motors and Generators" for electric motors. Include NEMA listing and labeling.
5. IEC 60034 - Rotating Electrical Machines.
6. NFPA 70 "National Electrical Code" for electrical components and installation.

B. Single-Source Responsibility: Obtain each category of pumps from 1 source and by a single manufacturer.

1.06 PERFORMACE GAURANTEE

A. Factory Performance Test:

1. The pumps manufacturer shall be responsible for conducting pump witness of performace test to validate the head-flow curves, pump efficiency curve, motor efficiency curve, NPSHr curve and vibration of the pump motor assembly.
2. The test shall be conducted according to the The Hydraulic Institute’s new test standard, ANSI/HI14.6.

3. The pump factory test shall be performed in the presence of two Owner Representatives and the Engineer and all the expenses including transportation, business class airfare, hotel accommodation…etc shall be borne by the Contactor and included in the pump cost.
4. The calibration of all instrumentation shall be traceable to the National Institute of Standards and Technology (NIST) or equivalent.
5. A test method statement shall be forwarded to the Engineer and approval shall be granted one month prior the test.
6. A certified test report of all data and test result shall be forwarded to the Engineer for acceptance prior shipping. The report shall include calibration curves, test results, pump curves as tested and information sheets for all instrumentation.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Store pumps in dry location.

B. Retain shipping flange protective covers and protective coatings during storage.

C. Protect bearings and couplings against damage from sand, grit, and other foreign matter.

D. Extended Storage Longer than 15 Days: Dry internal parts with hot air or vacuum-producing device. Coat internal parts with light oil, kerosene, or antifreeze after drying. Dismantle bearings and couplings; dry; coat with acid-free, heavy oil; tag; and store in dry location.

E. Comply with pump manufacturer's rigging instructions.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated in the Work include are mentioned in the Approved Manufactueres List.

2.02 PUMPS, GENERAL

A. General: Factory assembled and tested.

B. Base-Mounted Pumps: Include pump casings that allow removal and replacement of impellers without disconnecting piping.

C. Types, Sizes, Capacities, and Characteristics: As indicated.

D. Motors: NEMA MG 1 / IEC 60034, general purpose, premium or super premium efficiency based on available sizes, continuous duty, Design B, except Design C where required for high starting torque. Furnish single-, or variable-speed motors, with type of enclosures and electrical characteristics indicated. Include built-in thermal-overload protection and grease-lubricated ball bearings. Select each motor to be nonoverloading over full range of pump performance curve.

Provide Active front end - low harmonic variable frequency drives (THDi <5) according to IEEE 519 as scheduled. Select motor with premium efficiency (IE-3). For more electrical details Refer to Section (23 05 13) & (26 24 19), COMMON ELECTRICAL REQUIREMENTS FOR HVAC EQUIPMENT & MOTOR CONTROL CENTERS.

E. Electrical Design Requirements

1. Starting torque ratings at 100% and 65% of design operating voltages shall be guaranteed.
2. The motor service factor shall be 1.15.
3. Stator insulation shall be NEMA class F.
4. Stator temperature rise shall be NEMA class B at the rated horsepower, and NEMA class F at the specified service factor.
5. The design ambient temperature shall be 50 °C (122 °F).
6. Motor thermal protection through PT100 sensor.
7. Motor space heater shall be included.
8. If driven by VFD, the motor should be inverter duty type.

F. Factory Finish: Manufacturer's standard epoxy paint applied to factory-assembled and -tested units before shipping. The pump, motor, base plate and frame color to be according to the engineer’s direction.

G. Manufacturer's Preparation for Shipping: Clean flanges and exposed machined metal surfaces and treat with anticorrosion compound after assembly and testing. Protect flanges, pipe openings, and nozzles with wooden flange covers or with screwed-in plugs.

H. Pumps shall be designed using hydraulic criteria based upon actual model developmental test data. Manufacturer shall certify that pumps have been hydraulically tested at the factory.

I. Minimum pump efficiency shall be no less then 85 % for 150 KW or larger pumps, and no less then 80 % for all other smaller pumps.

J. Head-capacity curves shall slope up to maximum head at shut-off. Curves shall be relatively flat for closed systems. Select pumps near the midrange of the curve, so that the design capacity falls to the right of the best efficiency point for all variable speed pumps, without approaching the pump curve end point and possible cavitation and unstable operation Select pumps for open systems so that available net positive suction head (NPSHa) exceeds the net positive head required (NPSHr) by at least 1.5 times. Deviations within 3 percent of maximum efficiency are permissible, provided the lesser efficiency is not less than the specified efficiency

K. The head for pumps submitted for pumping through condensers and through chilled water coils and evaporators shall be increased, if necessary, to match the equipment approved for the project

L. Pump Driver: Furnish with pump. Size to be non-overloading at any point on the head-capacity curve including one pump operation in a parallel pumping installation.

M. For variable speed pumps, pumps shall be selected with the maximum impeller size to provide the duty point at no more than 85% of the full load rpm, such that there is a margin to increase the head or flow in case of any shortage.

N. In case of using constant speed pumps, the pump impeller diameter shall be selected around 90-95 % of the published maximum diameter of the casing

O. Acceptable maximum impeller diameter calculations shall not be based on percentage of impeller diameter range for a given casing. Shop Drawings will be approved only if complete performance curves for all impeller sizes for a given casing are included in the submittal.

P. Where parallel-pump operation is indicated, pumps selected shall have characteristics specifically suitable for the service without unstable operation.

Q. When net positive suction head (NPSH) calculations are made> inlet-storage head shall be considered at empty point and the centerline pump location shall include not less than a 150 mm concrete base.

R. Pumps of the same duty condition, classification, and accessories, or with specified accessory deviation, shall be identical and the product of one manufacturing source.

S. Furnish each pump and motor with a name plate giving manufacturer's name. serial number of pump, capacity in l/s and head in Kpa at design conditions, Power KW, voltage, frequency, speed, full load current, power factor and motor efficiency.

T. Pumps shall be capable of operating continuously without overheating bearings of motors at every condition of operation on the pump curve at each operational speed, or produce noise audible outside the room or space at which it is installed.

U. Pump casing pressure shall be higher than the maximum working pressure, and shall be factory tested at 11/2 times the design pressure.

V. Provide coupling gaurds to meet ANSI B15.1, section 8 and OSHA requirements.

W. Pumps shall be capable of operation and speed control as shown on drawings.

2.03 END SUCTION – FLEXIBLE COUPLED CENTRIFUGAL PUMP

A. Type: Horizontal, base mounted, end suction, single stage, centrifugal type, directly connected to motr through a heavy duty flexible coupling, with heavy gauge coupling guard. Pumps to be constant speed or variable speed type provided with variable speed drives as shown on drawings.

B. Base: Pump and motor are to be mounted on a common cast iron adequately reinforced against deflection, with drip rim, drain tapping, bolt holes and grouting hole.

C. Bearings: Pumps rotating element to be supported by two heavy duty grease lubricated ball bearings mounted in heavy iron frame with adequate supports to the base fo maximum rigidity.

D. Pump Casing: High tensile strength close grain cast iron with smooth waterways, register fitted and bolted to bearing frame for permanent alignment. It is to be fitted with bronze wear rings, and with tapped and plugged bottom drain and top vent connections.

E. Pump Casing shall be rated for a minimum working pressure of 1750 kPa.

F. Impeller: Bronze, enclosed type, fitted to shaft with a key and locked in place.

G. Shaft: One piece stainless steel, sized to carry axial and radial thrust with minimum deflection.

H. Mechanical Seal: Ni-resist face, carbon washer and stainless steel metal parts.

I. Electric Motor: Totally enclosed, fan cooled, squirred cage, induction type, with 1450 RPM permanently lubricated and sealed ball bearings.

J. Install a polyurethane abrasive separator or cyclone separator, in the seal piping with a sight type flow indicator, piped in accordance with the manufacturer's instructions.

K. Coupling Guard: Steel, removable, and attached to mounting frame.

L. Coupling: Flexible-spacer type, capable of absorbing torsional vibration and shaft misalignment for motor sizes of 100 hp and smaller; with flange and sleeve section that can be disassembled and removed without removing pump or motor, for sizes larger than 100 hp.

2.04 END SUCTION CLOSED COUPLED CENTRIFUGAL PUMP

A. Type: Horizontal, base mounted, end suction, single stage, centrifugal pump, fully bronze fitted, back-pull-out, radially split case design, closed coupled motor and pump shafts rated for a minimum working pressure of 1750 kPa. and a continuous working temperature of 107 deg C.
B. Wear Rings: Replaceable, bronze casing ring.

C. Bearings: Grease lubricated ball bearings

D. Pump Casing: Cast iron, with flanged piping connections, drain plug at low point of volute, and threaded gage tapings at inlet and outlet connections

E. Impeller: ASTM B 584, cast bronze, statically and dynamically balanced, closed overhung, single suction, keyed to shaft, and secured by locking cap screw.

F. Mechanical Seal: Mechanical, with carbon-steel rotating ring, stainless-steel spring, ceramic seat, and flexible bellows and gasket

G. Electric Motor: Directly mounted to pump casing and with supporting legs as integral part of motor enclosure.

H. Pump shaft: Stainless-steel shaft close coupled to motor shaft

2.05 AXIALLY SPLIT CASE PUMPS

A. Description: Mounting feet cast integrally with the lower half casing, centrifugal pump with motor, volute type, double-suction, single or multi-stage, bronze-fitted, axially split case design; rated for 1750 kPa minimum working pressure and a continuous water temperature of 107 deg C, with an impeller mounted between grease lubricated out-board bearings and with a flexible coupling connecting the motor and the pump shaft.

B. Casing: Close grained Cast iron, with ASME B16.1, Class 250 flanged pipe connections. Include threaded gage tapings at inlet and outlet flange connections, vent valve at top of volute, and threaded drain plug at bottom of volute.

C. Casing Vent: manual brass cock at high point and automatic air vent.

D. Casing Drain and Gauge Taps: minimum 25mm (1-inch) plugged connections.

C. Impeller: ASTM B 584 cast bronze with wearing rings, statically and dynamically balanced, enclosed, double inlet and keyed to shaft. Impeller removable without disturbing pipe connections or motor.

D. Wearing Rings: Replaceable, bronze casing ring.

E. Shaft and Sleeve: Stainless Steel shaft with bronze or stainless steel sleeve.

F. Pump Shaft Bearings: Grease-lubricated ball bearings contained in cast-iron housing. Provide lip seal and slinger outboard of each bearing.

G. Seals: Mechanical type. Include carbon-steel rotating ring, stainless-steel spring, ceramic seat, and flexible bellows and gasket.

H. Coupling: Flexible, capable of absorbing torsional vibration and shaft misalignment. Include flexible-spacer type, with flange and sleeve section that can be disassembled and removed without removing pump or motor.

I. Mounting Frame: Welded-steel frame and cross members, factory-fabricated from ASTM A 36 (ASTM A 36M) channels and angles. Fabricate for mounting pump casing, coupling guard, and motor. Grind welds smooth before application of factory finish. Field-drill motor-mounting holes for field-installed motors. Fabricated to be mounteded on a concrete base. Field or local assembling is not accepted.

J. Motor: Secured to mounting frame, with adjustable alignment.

K. Install an abrasive separator or cyclone separator, in the seal piping with a sight type flow indicator, piped in accordance with the manufacturer's instructions.

M. Casing and bearing housing shall be Epoxy coated at factory prior shipment.

2.06 HORIZONTALLY SPLIT CASE PUMPS

A. Manufacturers:

1. Subject to compliance with requirements, manufacturers offering products that may be incorporated in the work are indicated in project list of appoved manufacturers.
2. The contractor is to submit three manufacturers from the list and the client has the right to select one of them.

B. General:

3. Provide pumps that will operate continuously without overheating bearings or motors at every condition of operation on the pump curve, or produce noise audible outside the room or space in which installed.
4. Provide pumps of size, type and capacity as indicated, complete with electric motor and drive assembly, unless otherwise indicated. Design pump casings for the indicated working pressure and factory test at 1½ times the designed pressure.
5. Provide pumps of the same type, the product of a single manufacturer, with pump parts of the same size and type interchangeable.
6. General Construction Requirements

a. Balance: Rotating parts, statically and dynamically.

b. Construction: To permit servicing without breaking piping or motor connections.

c. Pump Motors: Provide premium or super premium efficiency motors, inverter duty for variable speed service. Motors shall be totally closed fan cooled and operate at 950 rpm or 1450 rpm unless noted otherwise. NEMA MG 1, general purpose, continues duty. Furnish variable speed motors, with type of enclosures and electrical characteristics indicated. Include built-in thermal-overload protection and grease-lubricated ball bearings. Select each motor to be non-overloading over full range of pump performance curve.

d. Provide coupling guards that meet ANSI B15.1, Section 8 and OSHA requirements.

e. Pump Connections: Flanged.

f. Pump shall be factory witnessed tested by the client and consultants to demonstrate full performance at specified conditions. Include for all the test charges. Cost shall include full board five stars accommodation, national airline business Class for minimum three persons.

g. Performance: As scheduled on the Contract Drawings.

5. Variable Speed Pumps:

a. The pumps shall be the type shown on the drawings and specified herein flex coupled to a totally-closed, fan-cooled motor.

b. All VFD are to be Active front end- low harmonic drive (THDi< 5%) According to IEEE 519. Harmonic calculation shall be provided upon request.

c. All VFDs to include input line filter sinus filter and motor output choke if applicable.

d. VFD to be sized bigger than motor rating at 100% fulload with a suitable marigin that allows for smooth operation.

e. VFD to be light load duty rated.

f. Internal EMC filter as well as a RFI filter (2^{nd} environment) are to be provided.

g. Input and output line reactor are to be provided.

h. VFD are to include space heater.

i. Protection functions of VFD: Over current, short circuit, input/output phase loss, motor overload and under load,over/under-voltage, over speed, over temperature, motor stall, Ground fault protection, motor thermal protection, variable torque overload protection,other internal fault.

j. Variable Speed Motor Controllers: Refer to Section (section 26 24 19) - MOTOR CONTROL CENTERS.

k. . Furnish controllers with pumps and motors.

l. Pump operation and speed control shall be as shown on the drawings.

C. Double Suction Type:

1. Casing and Bearing Housing: Close-grained cast iron, ASTM A48. Manufacturer's Epoxy coat applied internally and externally, to be factory-assembled and factory-tested before shipping. Internal polishing of casing is mandatory.
2. Casing Wear Rings: Bronze.
3. Suction and Discharge: Plain face flange, 850 kPa (125 psig), ANSI B16.1.
4. Casing Vent: Manual brass cock at high point and automatic air vent.
5. Casing Drain and Gage Taps: minimum 25 mm (1-inch) plugged connections minimum size.
6. Impeller: Bronze, ASTM B62, enclosed type, keyed to shaft. Impeller diameter shall be the maximum diameter allowed and giving the best efficiency for variable speed application.
7. Shaft: stainless steel with bronze sleeve.
8. Shaft Seal: Manufacturer's standard mechanical type to suit pressure and temperature and fluid pumped. Include carbon-steel rotating ring, stainless steel spring, ceramic seat, and flexible bellows and gasket. Install an abrasive separator, in the seal piping with a sight type flow indicator, piped in accordance with the manufacturer's instructions.
9. Shaft Sleeve: Bronze or stainless steel.
10. Where parallel-pump operation is indicated, pumps selected shall have characteristics especially suitable for the service without unstable operation.
11. Motor: Furnish with pump. Refer to Section 23 05 12.
12. Designed for disassembling for service or repair without disturbing the piping or removing the motor.
13. Impeller Wear Rings: Bronze.
14. Shaft Coupling: Non-lubricated steel flexible type or spacer type with coupling guard, ANSI B15.1, bolted to the base plate.
15. Bearings (Double-Suction pumps): Regreaseable ball or roller type. Provide lip seal and slinger outboard of each bearing.

E. Base: Cast iron or fabricated steel for common mounting to a concrete base, factory designed, fabricated and assembled. Field or local assembling is not accepted.

F. Provide line sized shut-off valve and suction strainer, maintain manufacturer recommended straight pipe length on pump suction (with blow down valve). Contractor option: Provide suction diffuser as follows:

1. Body: Cast iron with steel inlet vanes and combination diffuser-strainer-orifice cylinder with 5 mm (3/16-inch) diameter openings for pump protection. Provide taps for strainer blow down and gage connections.
2. Provide adjustable foot support for suction piping.
3. Strainer free area: Not less than five times the suction piping.
4. Provide disposable start-up strainer.

2.07 IN-LINE CIRCULATORS

A. Description: , Centrifugal pump with motor, volute type, single suction in-line, single-stage, fully bronze-fitted, radially split case design; oil lubricated bearings, rated for 1750 kPa minimum working pressure and a continuous water temperature of 107 deg C.

B. Casing: Cast iron, with threaded companion flanges for piping connections smaller than 65 mm and threaded gage tapings at inlet and outlet connections.

C. Connection Option: Include unions, instead of threaded companion flanges, at connections for casings that are not available with threaded companion flanges.

D. Impeller: ASTM B 584, cast bronze, statically and dynamically balanced, closed, overhung, single suction, and keyed to shaft. Impeller removable should be without disturbing pipe connections

E. Shaft and Sleeve: Stainless Steel shaft with oil-lubricated copper sleeve.

F. Seals: Mechanical type. Include carbon-steel rotating ring, stainless-steel spring, ceramic seat, and flexible bellows and gasket.

G. Pump Bearings: Oil-lubricated, bronze journal and thrust type.

H. Motor Bearings: Oil-lubricated, sleeve type.

I. Coupling: Flexible, capable of absorbing torsional vibration and shaft misalignment.

J. Motor: Resiliently mounted to pump casing.

2.08 COMPACT IN-LINE CIRCULATORS

A. Compact in-line circulators are inexpensive solutions for low-flow and low-head applications, the following are the main types:

1. Water cooled type, horizontal, in-line, compact design, seal-less, centrifugal, and single stage. Include pump and motor assembled on a common shaft in hermetically sealed unit, without stuffing boxes or mechanical seals. Include lubrication of sleeve bearing and cooling of motor by circulating pumped liquid through motor section, and isolation of motor section from motor-stator windings by corrosion-resistant, nonmagnetic, alloy liner. Include design rated for 860-kPa minimum working pressure and a continuous water temperature of 107 deg C.
2. Cartridge type, horizontal, in-line, compact, seal-less, centrifugal, and single stage. Include pump and motor assembled on a common shaft in cartridge-type, hermetically sealed unit, without stuffing boxes or mechanical seals. Include isolation of motor section from motor-stator windings by corrosion-resistant, nonmagnetic, alloy liner. Include design rated for 1750-kPa minimum working pressure and a continuous water temperature of 107 deg C.

B. Casing: Cast bronze, with stainless-steel liner, static O-ring seal to separate motor section from motor stator, and flanged piping connections.

C. Impeller: Overhung, single suction, closed or open, non-metallic.

D. Shaft and Sleeve: Stainless-steel shaft with carbon-steel sleeve.

E. Motor: Single or variable-speed.

2.09 VERTICAL IN-LINE PUMPS

A. Centrifugal, flexible-coupled, single-stage radially split case design. Include vertical-mounting, bronze-fitted design and mechanical seals rated for 1750-kPa minimum working pressure and a continuous water temperature of 107 deg C.

B. Casing: Cast iron, with threaded companion flanges for piping connections smaller than DN 80, drain plug at low point of volute, and threaded gage tapings at inlet and outlet connections.

C. Impeller: ASTM B 584, cast bronze, statically and dynamically balanced, closed, overhung, single suction, and keyed to shaft.

D. Shaft and Sleeve: Ground and polished stainless-steel shaft with bronze sleeve.

E. Seals: Mechanical, with carbon-steel rotating ring, stainless-steel spring, ceramic seat, and flexible bellows and gasket.

F. Motor: Directly mounted to pump casing and with lifting and supporting lugs in top of motor enclosure.

2.10 PUMP SPECIALTY FITTINGS

A. Provide line sized shut-off valve and suction strainer, maintain manufacturer recommended straight pipe length on pump suction (with blow down valve). Contractor option: Provide suction diffuser as follows:

1. Body: Cast iron with bronze or stainless-steel straightening inlet vanes. Complete with bronze startup and bronze or stainless-steel permanent strainers (combination diffuserstrainer- orifice cylinder with 5 mm (3/16-inch) diameter openings for pump protection). Angle or straight pattern, having system pressure rating or higher.; Provide taps for strainer blow down and gage connections.
2. Provide adjustable foot support for suction piping.
3. Strainer free area: Not less than five times the suction piping.
4. Provide disposable start-up strainer.

B. Triple-Duty Valve: Angle or straight pattern, 1750-kPa pressure rating, cast-iron body, pump-discharge fitting; with drain plug and bronze-fitted shutoff, balancing, and check valve features.

2.11 GENERAL-DUTY VALVES

A. Refer to Division 23 Section 230523 "General- Duty Valves for HVAC Piping" for general-duty gate, ball, butterfly, globe, and check valves.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine areas, equipment foundations, and conditions, with Installer present, for compliance with requirements for installation and other conditions affecting performance of pumps.

B. Examine roughing-in for piping systems to verify actual locations of piping connections before pump installation.

C. Examine foundations and inertia bases for suitable conditions where pumps are to be installed.

D. Do not proceed until unsatisfactory conditions have been corrected.

3.02 INSTALLATION

A. Follow manufacturer's written instructions for pump mounting and start-up. Access/Service space around pumps shall not be less than minimum space recommended by pumps manufacturer.

B. Install pumps in locations indicated and arranged to provide access for periodic maintenance, including removal of motors, impellers, couplings and accessories.

C. Contractor shall provide loading and unloading facility for equipment handling in the pump room. Permanent monorail with motorized hoist and with remote operation is minimum requirement.

D. Support pumps and piping separately so that piping is not supported by pumps.

E. Provide drains for bases and seals for base mounted pumps, piped to and discharging into floor drains through funnel.

F. Install non-slam non return valve on the discharge of the pump

G. Keep the recommended straight distance upstream and downstream of all equipment as directed by the respective manufacturers.

H. Coordinate location of thermometer and pressure gauges as per Section 23 21 13, HYDRONIC PIPING.

I. Install differential pressure indication transducer across the pump suction and discharge as well for the suction strainer.

J. Set base-mounted pumps on concrete foundation. Disconnect coupling halves before setting. Do not reconnect couplings until alignment operations have been completed.

1. Support pump base plate on rectangular metal blocks and shims, or on metal wedges with small taper, at points near foundation bolts to provide a gap of 19 mm to 38 mm between pump base and foundation for grouting.

K. Provide VFD on pumps if indicated on drawings.

L. Contractor shall provide and grout the base of all pumps as recommended by the pump manufacturer.

M. Contractor shall provide a 3-plane vibration test for all pumps 20 HP and larger. Contractor shall submit three (3) copies of test results to the Engineer for review. All vibration testing shall be provided by a third party, Certified in 3 plane vibration testing.

3.03 ALIGNMENT

A. Align pump and motor shafts and piping connections after setting them on foundations, after grout has been set and foundation bolts have been tightened, and after piping connections have been made.

B. Comply with pump and coupling manufacturers' written instructions.

C. Adjust alignment of pump and motor shafts for angular and parallel alignment by 1 of 2 methods specified in the H.I.'s Standards for Centrifugal, Rotary & Reciprocating Pumps, "Instructions for Installation, Operation and Maintenance".

D. After alignment is correct, tighten foundation bolts evenly but not too firmly. Fill base plate completely with nonshrink, nonmetallic grout, with metal blocks and shims or wedges in place. After grout has cured, fully tighten foundation bolts.

E. Alignment Tolerances: According to manufacturer's recommendations.

3.04 CONNECTIONS

A. Piping installation requirements are specified in other Division 23 Sections. Drawings indicate general arrangement of piping, fittings, and specialties. Install shutoff valve and strainer on pump suction and check valve and shutoff valve on pump discharge, except where other arrangement is indicated.

B. Install piping adjacent to machine to allow service and maintenance.

C. Connect piping to pumps. Install valves that are the same size as piping connecting to pumps.

D. Install suction and discharge pipe sizes equal to or greater than the diameter of pump nozzles.

E. Install thermometers where indicated.

F. Install suction diffuser and shut off valve on the suction side of pumps.

G. Install flexible connectors (double sphere or high flexibility type based on the recommendation from the stress analysis study) on suction and discharge sides of base-mounted pumps and where indicated. Install between pump casing and valves, except where other arrangement is indicated.

H. Install pressure gages on pump suction and discharge, at integral pressure gage tappings, or install gauge with multiple input selector vavle.

I. Install electrical connections for power, controls, and devices.

J. Electrical power and control wiring and connections are specified in Division 26 Sections.

K. Ground equipment according to Division 26 Section "Grounding and Bonding for Electrical

Systems."

L. Connect wiring according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."

3.05 FIELD QUALITY CONTROL

A. Check suction piping connections for tightness to avoid drawing air into pumps.

B. Clean strainers.

C. Set pump controls.

3.06 COMMISSIONING

A. Final Checks Before Startup: Perform the following preventive maintenance operations and checks before startup:

1. Lubricate bearings.
2. Remove grease-lubricated bearing covers, flush bearings with kerosene, and clean thoroughly. Fill with new lubricant according to manufacturer's recommendations.
3. Disconnect coupling and check motor for proper rotation that matches direction marked on pump casing.
4. Check that pumps are free to rotate by hand. Pumps for handling hot liquids shall be free to rotate with pump hot and cold. Do not operate pump if it is bound or even drags slightly until cause of trouble is determined and corrected.
5. Check that pump controls are correct for required application.

B. Starting procedure for pumps with shutoff power not exceeding safe motor power:

1. Prime pumps, opening suction valve, closing drains, and preparing pumps for operation.
2. Start motors.
3. Open discharge valves slowly.
4. Check general mechanical operation of pumps and motors.

B. Refer to volume 23 Section "Testing, Adjusting, and Balancing" for detailed requirements for testing, adjusting, and balancing hydronic systems.

3.7 START-UP:

A. Contractor shall provide start-up service from the Manufacturer's Representative, which shall include the alignment of the pump coupling for all pumps. Laser alignment shall be completed for pumps 20 HP and larger.

B. Start-up Certificate: Submit the start-up certificate to the Architect/Engineer within seven (7) days from the day of start-up. A copy of the start-up certificate shall be available on-site, the day the start-up is complete, and remain available at the site for the duration of the project.

1. Verify that the piping system has been flushed, cleaned and filled.
2. Lubricate pumps before start-up.
3. Prime the pump, vent all air from the casing and verify that the rotation is correct. To avoid damage to mechanical seals, never start or run the pump in dry condition.
4. Verify that correct size heaters-motor over-load devices are installed for each pump controller unit.
5. Field modifications to the bearings and or impeller (including trimming) are not permitted. If the pump does not meet the specified vibration tolerance ; send the pump back to the manufacturer for a replacement pump. All modifications to the pump shall be performed at the factory.
6. Ensure the disposable strainer is free of debris prior to testing and balancing of the Hydronic system.
7. After several days of operation, replace the disposable start-up strainer with a regular strainer in the suction diffuser.
8. Pumps shall not be used for flushing purpose.

3.8 DEMONSTRATION:

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain Hydronic pumps. Refer to Division 01 Section "Demonstration and Training."

END OF SECTION 23 21 23